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Electromagnetic

Assignment 1

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1 Problem 1

First we calculate vectors in the given point in space. And then the perpendicular vector is determined by cross product of $\mathbf{A}$ and $\mathbf{B}$ at that point.

$$\mathbf{A} = 4\hat{r} - \sqrt{2}\hat{\theta} + \hat{\phi} \quad \mathbf{B} = \sqrt{2}\hat{\theta} + \hat{\phi} \quad (1)$$

$$\frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|} = -0.378\hat{r} - 0.534\hat{\theta} + 0.756\hat{\phi} \quad (2)$$

2 Problem 2

$$\int_C \sin(\varphi) \hat{a}_r d\varphi \quad (3)$$

For take this integral we need to write the function in x-y coordinate system.

$$\mathbf{A} = \sin \varphi \cos \varphi \hat{i} + \sin \varphi \sin \varphi \hat{j} \quad (4)$$

$$\int_C \mathbf{A} d\varphi = \frac{\pi a}{2} \hat{j} \quad (5)$$

3 Problem 3

As definition for a counter-clockwise path we define $d\mathbf{l}$ as following.

$$d\mathbf{l} = -a \sin \varphi d\varphi \hat{i} + a \cos \varphi d\varphi \hat{j} \quad (6)$$

$$\int_C V d\mathbf{l} = a^2 \int_0^\pi \cos^2 \varphi \left(-\sin \varphi \hat{i} + \cos \varphi \hat{j} \right) d\varphi = a^2 \quad (7)$$

$$\int_C V d\mathbf{l} = -\frac{2}{3}a^2 \hat{i} \quad (8)$$

4 Problem 4

To determine the integral, we divide the surface into two. One is part of a sphere. The other one, is body of a cone.

$$\hat{n}_1 = \hat{r} \quad , \quad \hat{n}_2 = \hat{\theta} \quad (9)$$

$$\int_S \mathbf{A} \cdot d\mathbf{S} = \int_{S_1} \mathbf{A} \cdot \hat{n}_1 a^2 \sin \theta \, d\theta d\varphi + \int_{S_2} \mathbf{A} \cdot \hat{n}_2 r \sin \theta \, dr d\varphi \quad (10)$$

$$I_1 = \int_{\theta=0}^{\frac{\pi}{6}} \int_{\varphi=0}^{2\pi} a^4 \sin \theta^2 \, d\varphi d\theta = a^4 \pi \left(\frac{\pi}{6} - \frac{\sqrt{3}}{4} \right) \quad (11)$$

$$I_2 = \int_{R=0}^a \int_{\varphi=0}^{2\pi} 4r^3 \cos \frac{\pi}{6} \sin \frac{\pi}{6} \, dr d\varphi = a^4 \pi \frac{\sqrt{3}}{2} \quad (12)$$

$$\int_S \mathbf{A} \cdot d\mathbf{S} = I_1 + I_2 = a^4 \pi \left(\frac{\pi}{6} + \frac{\sqrt{3}}{4} \right) \quad (13)$$

5 Problem 5

5.1 a)

$$\nabla V = e^{7x+16y} \left(7 \cos 4z \, \hat{i} + 16 \cos 4z \, \hat{j} - 4 \sin 4z \, \hat{k} \right) \quad (14)$$

$$\nabla V(\mathbf{P}) = 2517 \, \hat{i} + 5753 \, \hat{j} - 3700 \, \hat{k} \quad (15)$$

5.2 b)

$$\nabla C = 4e^{-3z} (\sin \varphi \, \hat{r} + \cos \varphi \, \hat{\varphi} - 3r \sin \varphi \, \hat{z}) \quad (16)$$

$$\nabla C(\mathbf{P}) = 2 \, \hat{r} + 2\sqrt{3} \, \hat{\varphi} - 18 \, \hat{z} \quad (17)$$

5.3 c)

$$\nabla S = \frac{1}{R^5} \left(-4 \sin \theta \sin \varphi \, \hat{r} + \cos \theta \sin \varphi \, \hat{\theta} - \cos \varphi \, \hat{\varphi} \right) \quad (18)$$

$$\nabla S(\mathbf{P}) = -\sqrt{3} \, \hat{r} + \frac{3}{4} \, \hat{\theta} - \frac{1}{2} \, \hat{\varphi} \quad (19)$$